

Supramolecular Chemistry

Pathway Engineering in Pd-Based Supramolecular Cage Synthesis via Inner-Outer Steric Synergy

Ziteng Guo, Hao Yu, Junjuan Shi, Ningxu Han, Guanglu Wu, Houyu Zhang, Bingling Li, and Ming Wang*

Abstract: For artificial supramolecular architectures designed to mimic biological systems, achieving different pathway synthesis is challenging due to the requirement of multiple stable and interconvertible intermediates. Here, we propose a novel “inner-outer steric synergy” strategy and investigate controllable pathway engineering for the synthesis of specific structures. Firstly, three structures (**Ring-Pd₂L^A₂**, **Bowl-Pd₂L^A₃** and **Cage-Pd₂L^A₄**) with interconversion properties were selectively formed by assembling externally modified ligand **L^A** with Pd(II). Furthermore, **Ring-Pd₂L^A₂** can further assemble with the ligand **L^B** with inner steric hindrance to generate heteroleptic **trans-Pd₂L^A₂L^B₂** cage, while **Bowl-Pd₂L^A₃**, as an intermediate, can assemble with **L^B** to form **Pd₂L^A₃L^B**. It is noteworthy that **Ring-Pd₂L^A₂**, **Bowl-Pd₂L^A₃**, and **Cage-Pd₂L^A₄** can interconvert under specific conditions, enabling the synthesis of **Pd₂L^A₃L^B** and **trans-Pd₂L^A₂L^B₂** through 10 and 16 pathways, respectively. This research not only introduces a novel strategy for constructing heteroleptic cages but also demonstrates the achievement of pathway engineering.

Introduction

Inspired by nature's ingenious self-assembly, chemists have extensively studied various non-covalent interactions involved in self-assembly processes.^[1] Among them, metal coordination interactions have garnered significant attention due to their appropriate bond energies, high directionality and predictability.^[2] By harnessing these coordination interactions, a large number of precise discrete structures, such as fractal structures,^[3] rings,^[4] metal-organic cages,^[5] and helicates,^[6] have been successfully constructed. Notably, the

construction of these complex structures has bypassed traditional cumbersome chemical synthesis pathways, instead adopting a self-assembly strategy that can be achieved with simple ligands, thereby ensuring quantitative yields of the products. For example, Fujita successfully constructed a giant molecular sphere consisting of 144 components using Pd(II) and simple bis-pyridyl ligands.^[7] Additionally, Clever, Crowley, Lewis, and others have synthesized low-symmetry cages through the assembly of heteroligands, heterometals, or dissymmetrical ligands.^[8] These supramolecular architectures not only have unique structures, but also exhibit broad application potential in areas such as drug delivery,^[9] selective catalysis,^[10] and chiral recognition.^[11]

Pathway engineering, a concept that has garnered increasing attention in various fields, refers to the strategic design and control of reaction pathways to direct the synthesis of specific products or structures.^[12] This approach is widely used in chemical synthesis, materials science, and even biological systems to optimize and guide complex processes.^[13] In supramolecular chemistry, pathway engineering not only offers novel and effective strategies for constructing supramolecular structures, but also provides alternative solutions when specific synthetic pathways face obstacles, ensuring the successful formation of target products. Despite its significance, reports on pathway engineering in supramolecular assembly are still limited.^[14] This is because the achievement of pathway engineering of supramolecules requires the following two conditions: (a) the assembly process needs to be completed through intermediates; (b) these intermediates can interconvert with each other. Based on these two prerequisites, we have constructed a model (Figure 1a). However, in traditional self-assembly processes, capturing and utilizing these crucial intermediates is often challenging due to their insufficient stability.^[15] Meanwhile, the mutual transformation process between different structures tends to produce mixtures.^[16] To avoid this problem, careful design of the ligands is imperative.

In this study, we propose an inner-outer steric synergy strategy (named IOSS) for achieving the pathway engineering in the synthesis of heteroleptic cages. We designed and synthesized a ligand, **L^A** featuring outer modification, alongside another ligand, **L^B**, incorporating inner modification (Figure 1b). **L^A** exhibited good tunability in the assembly with Pd(II) and could controllably form three structures: **Ring-Pd₂L^A₂**, **Bowl-Pd₂L^A₃** or **Cage-Pd₂L^A₄** by altering the assembly conditions. Moreover, these three structures can undergo mutual transformation under certain conditions.

[*] Z. Guo, H. Yu, J. Shi, N. Han, Prof. G. Wu, Prof. H. Zhang, Prof. M. Wang
 State Key Laboratory of Supramolecular Structure and Materials,
 College of Chemistry
 Jilin University
 Changchun, Jilin, 130012 (China)
 E-mail: mingwang358@jlu.edu.cn
 Prof. B. Li
 State Key Laboratory of Electroanalytical Chemistry
 Changchun Institute of Applied Chemistry, Chinese Academy of Sciences
 Changchun, Jilin 130022 (China)

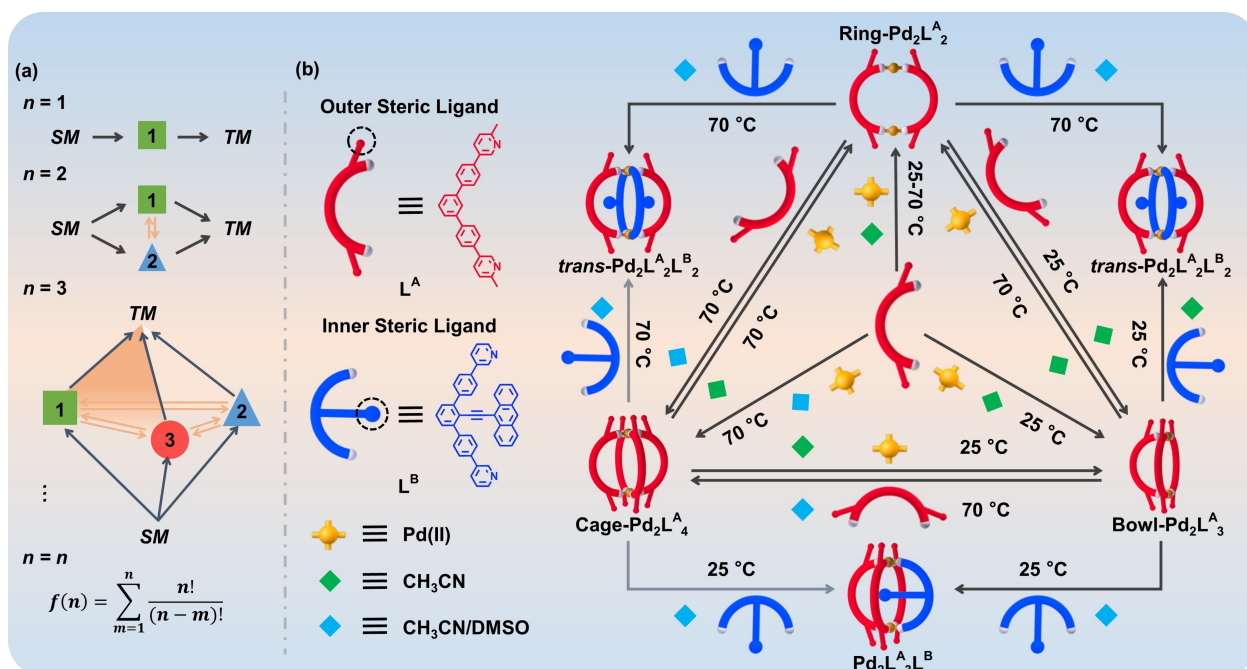


Figure 1. (a) Schematic diagram of multi-pathway synthesis (n = the number of intermediates; SM = starting material; TM = target molecule). (b) The synthesis of Ring-Pd₂L^A₂, Bowl-Pd₂L^A₃, Cage-Pd₂L^A₄, Pd₂L^A₃L^B and trans-Pd₂L^A₂L^B₂ (The light gray line represents partial conversion).

Due to the large steric hindrance effect, L^B cannot form a single discrete structure and tends to co-assemble with L^A. Through the synergy of inner and outer steric hindrance effects, utilizing Ring-Pd₂L^A₂ and Bowl-Pd₂L^A₃ as intermediates, the introduction of ligand L^B enables precise construction of two distinct heteroleptic cages, namely Pd₂L^A₃L^B and trans-Pd₂L^A₂L^B₂. Notably, based on the mutual transformation between the three L^A-based assembly structures, we have achieved the pathway engineering of supramolecular cages Pd₂L^A₃L^B and trans-Pd₂L^A₂L^B₂. Specifically, the Pd₂L^A₃L^B can be synthesized through 10 pathways, while the pathways to obtain trans-Pd₂L^A₂L^B₂ can be through up to 16 (Figure 1b).

Results and Discussion

The ligands L^A, L^B, and the reference ligand L^C, without additional substituents, were synthesized via efficient Suzuki and Sonogashira coupling reactions. Details of the synthesis and structural characterization were provided in the Supporting Information (Schemes S1–S3 and Figures S1–S14). Initially, assembly experiments were conducted with ligand L^A, Pd(II) under different conditions. Specifically, within the temperature range of room temperature to 70 °C, when L^A was assembled with Pd(CH₃CN)₄(BF₄)₂ in a 1:1 molar ratio in CD₃CN for 8 hours, only one set of proton signals was observed in the ¹H NMR spectrum (Figure 2a). DOSY spectroscopy revealed that the observed signals originated from a single species (Figure S62). Electrospray ionization mass spectrometry (ESI-MS) revealed a complex consisting of two L^A ligands and two Pd(II) ions (Figure S17).

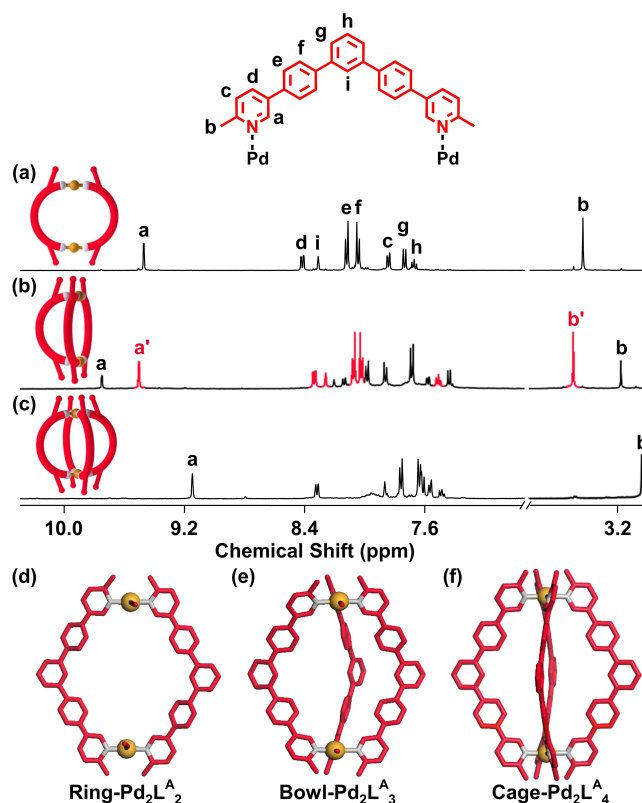


Figure 2. ¹H NMR spectra (500 MHz, 300 K) of (a) Ring-Pd₂L^A₂, (b) Bowl-Pd₂L^A₃ and (c) Cage-Pd₂L^A₄. The crystal structures of (d) Ring-Pd₂L^A₂ and (e) Bowl-Pd₂L^A₃. (f) DFT-optimized structure of Cage-Pd₂L^A₄ (Hydrogen atoms are omitted for clarity).

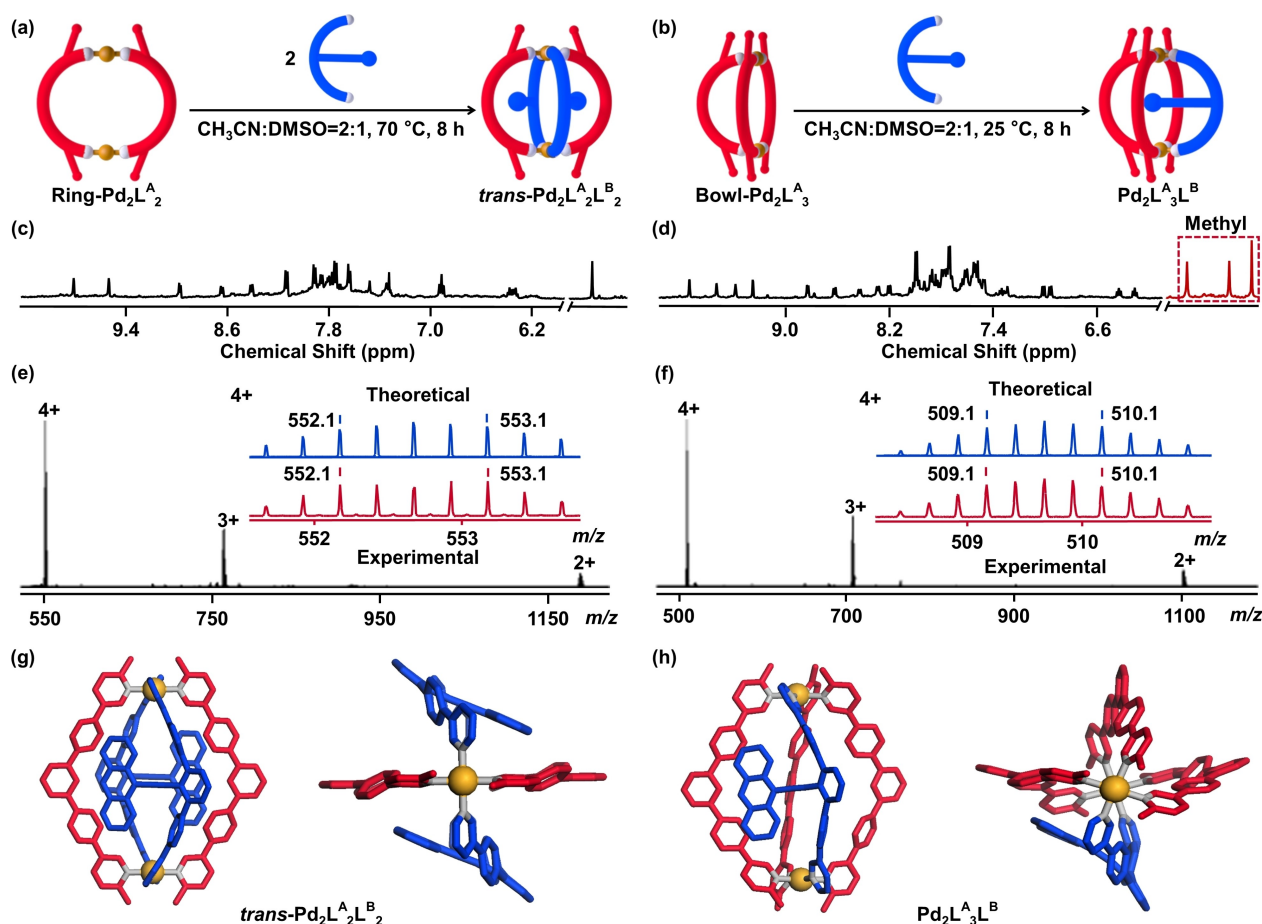


Figure 3. Assembly of (a) $\text{trans-Pd}_2\text{L}_2\text{L}_2^{\text{B}}_2$, (b) $\text{Pd}_2\text{L}_3\text{L}_3^{\text{B}}$. ^1H NMR spectra (500 MHz, 300 K) of (c) $\text{trans-Pd}_2\text{L}_2\text{L}_2^{\text{B}}_2$ and (d) $\text{Pd}_2\text{L}_3\text{L}_3^{\text{B}}$. ESI-MS spectra of (e) $\text{trans-Pd}_2\text{L}_2\text{L}_2^{\text{B}}_2$ and (f) $\text{Pd}_2\text{L}_3\text{L}_3^{\text{B}}$. (g) The front view (left) and top view (right) of the crystal structure of $\text{trans-Pd}_2\text{L}_2\text{L}_2^{\text{B}}_2$. (h) The front view (left) and top view (right) of the crystal structure of $\text{Pd}_2\text{L}_3\text{L}_3^{\text{B}}$ (Hydrogen atoms are omitted for clarity).

Furthermore, to further investigate the structural details, single crystals were successfully obtained by slowly evaporating tetrahydrofuran into the acetonitrile solution of the complex. Upon analysis, the complex exhibited a cyclic structure, with the two L^{A} ligands arranged in a *trans* configuration, while the remaining four coordination sites were occupied by CH_3CN (Figure 2d). Additionally, within the cyclic structure, there were two pairs of hydrogen atoms on its four methyl groups that were positioned closer than twice the van der Waals radius of hydrogen ($1.2 \text{ \AA}$), indicating repulsive forces between the methyl groups (Figure S78a).^[17] Through comprehensive NMR, MS, and single crystal characterization, we confirmed the complex to be a cyclic structure, termed **Ring-Pd₂L₂L₂^B₂**.

When the ratio of L^{A} to Pd(II) was adjusted to 3:2 and stirred at room temperature for 4 hours, no signal for **Ring-Pd₂L₂L₂^B₂** was detected in the ^1H NMR spectrum. Instead, two new sets of proton peaks with an integral ratio of 2:1 were observed (Figures 2b, S19), indicating that the ligand L^{A} is present in two different chemical environments, with a ratio of 2:1 between them. DOSY spectroscopy further confirmed that these signals originated from a single species (Figure S63). ESI-MS revealed the signal of $[\text{Pd}_2(\text{L}^{\text{A}})_3]^{4+}$

(Figure S21). The single-crystal structure revealed that the dinuclear complex adopts a bowl-like shape, termed **Bowl-Pd₂L₃L₃^B**, and possesses two distinct chemical environments for L^{A} , which is in good agreement with the ^1H NMR (Figure 2e). Interestingly, the remaining two coordination sites are occupied by H_2O , rather than CH_3CN , likely due to trace amounts of H_2O present in the solvent. Additionally, in **Bowl-Pd₂L₃L₃^B**, four pairs of hydrogen atoms on the six methyl groups are positioned closer than twice the van der Waals radius of hydrogen (Figure S78b). Despite stronger repulsive interactions between the methyl groups compared to **Ring-Pd₂L₂L₂^B₂**, **Bowl-Pd₂L₃L₃^B** proved to be stable for at least 60 days during which no changes were observed in the ^1H NMR spectrum (Figure S58). When the ratio of L^{A} to Pd(II) was changed to 2:1 and stirred in a mixed solvent ($\text{CD}_3\text{CN}/\text{DMSO}-d_6$, v/v 2:1) at 70°C for 8 hours, proton peak signals distinct from both **Ring-Pd₂L₂L₂^B₂** and **Bowl-Pd₂L₃L₃^B** were observed in the ^1H NMR spectrum (Figure 2c). DOSY spectroscopy confirmed that the observed signals originated from a single species (Figure S64). ESI-MS revealed a complex corresponding to four L^{A} and two Pd(II) ions, confirming the formation of **Cage-Pd₂L₄L₄^B** (Figure S25). Despite multiple attempts, obtaining suitable single crystals

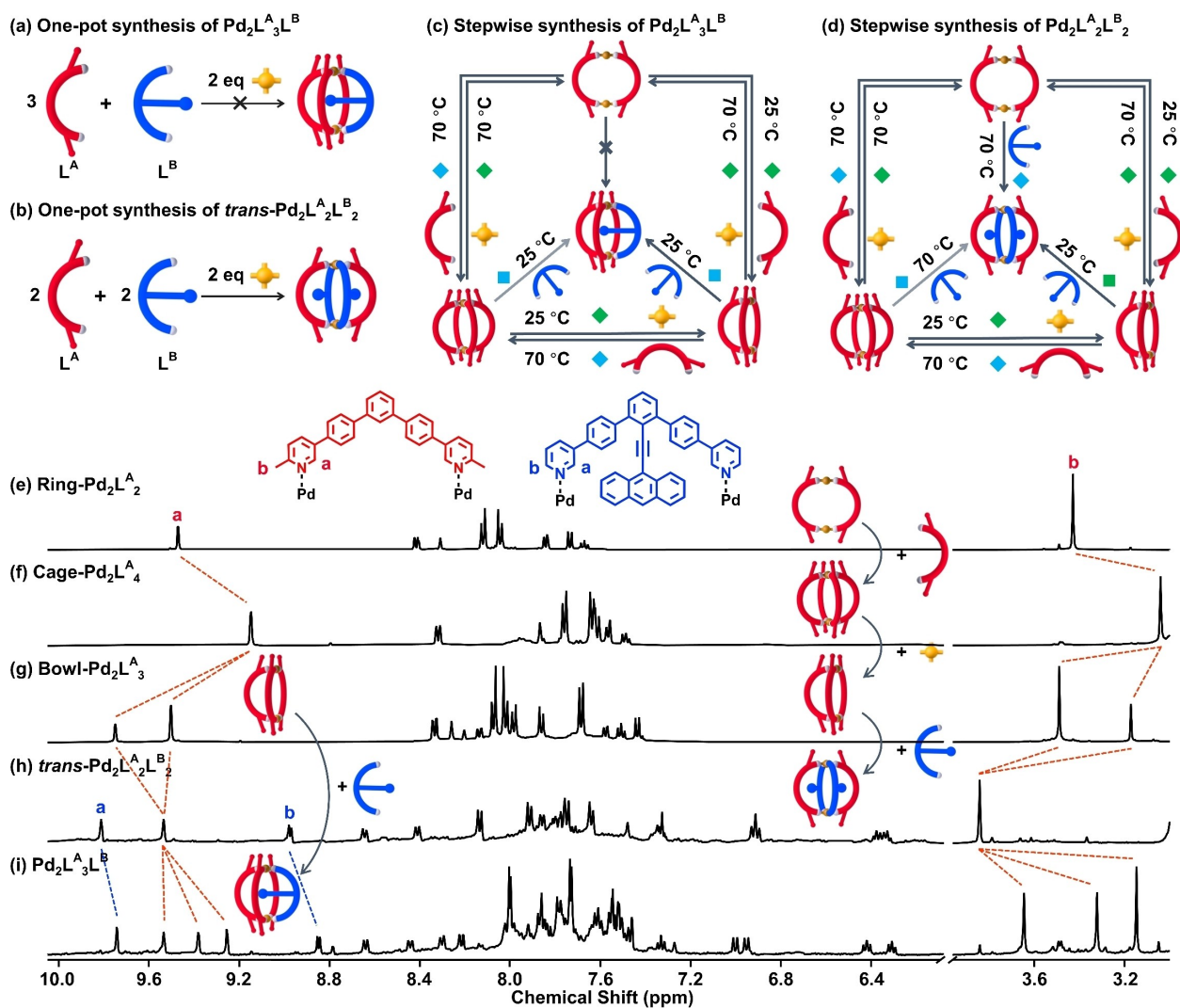


Figure 4. One-pot synthesis of (a) $\text{Pd}_2\text{L}_3\text{L}^{\text{A}}\text{L}^{\text{B}}$ and (b) $\text{trans-Pd}_2\text{L}_2\text{L}^{\text{A}}\text{L}^{\text{B}}$. The synthetic pathway diagrams for (c) $\text{Pd}_2\text{L}_3\text{L}^{\text{A}}\text{L}^{\text{B}}$ and (d) $\text{trans-Pd}_2\text{L}_2\text{L}^{\text{A}}\text{L}^{\text{B}}$. (e) ^1H NMR spectrum (500 MHz, 300 K) of **Ring-Pd₂L₂A₂** dissolved in CD_3CN . (f) ^1H NMR spectrum (500 MHz, 300 K) of **Ring-Pd₂L₂A₂** (dissolved in CD_3CN) after adding the ligand L^{A} (dissolved in $\text{DMSO}-d_6$). (g) ^1H NMR spectrum (500 MHz, 300 K) after the completion of the reaction between **Cage-Pd₂L₂A₂** and $\text{Pd}(\text{CH}_3\text{CN})_4(\text{BF}_4)_2$ in CD_3CN . (h) ^1H NMR spectrum (500 MHz, CD_3CN , 300 K) after adding ligand L^{B} to **Bowl-Pd₂L₂A₂**. (i) ^1H NMR spectrum (500 MHz, 300 K) after adding ligand L^{B} (dissolved in $\text{DMSO}-d_6$) to **Bowl-Pd₂L₂A₂** (dissolved in CD_3CN).

for testing were unsuccessful. Consequently, DFT geometric optimization was employed to simulate the structure of **Cage-Pd₂L₂A₂** (Figure 2f). Within the simulated structure, 12 pairs of hydrogen atoms with distances less than 2.4 Å were identified (Figure S79b). To quantitatively investigate the reasons behind the selectivity, we calculated the Gibbs free energy changes (ΔG) for the formation of two other potential complexes under the ligand-to-metal ratios required for the formation of three specific complexes. The calculation results indicate that, under the formation conditions of each complex, the thermodynamic driving force of the target complex is significantly higher than that of the other two potential complexes (Figures S83–S85).

Due to the remaining coordination sites in both **Ring-Pd₂L₂A₂** and **Bowl-Pd₂L₂A₂**, we initially considered introducing L^{B} into these two structures to construct heteroleptic cages. Compared with ligand L^{A} , the bulky ethynyl anthracene

group located in the concave region of ligand L^{B} was expected to reside inside the cage, without affecting the metal coordination center.^[18] Due to the large steric hindrance effect, L^{B} itself cannot be assembled into a single discrete structure (Figure S27). Therefore, we predicted that L^{B} would preferentially further assemble with **Ring-Pd₂L₂A₂** or **Bowl-Pd₂L₂A₂** to form heteroleptic cages. Firstly, for a CD_3CN solution of **Ring-Pd₂L₂A₂**, 2 equivalents of L^{B} (dissolved in $\text{DMSO}-d_6$) were added (Figure 3a). The reaction was carried out at 70 °C for 8 hours, and the ^1H NMR spectrum showed distinct new set of proton signals that were different from **Ring-Pd₂L₂A₂** (Figure 3c), indicating the formation of a new product. DOSY spectroscopy confirmed that these signals originated from a single species (Figure S65). The ESI-MS spectrum displayed a series of signal peaks corresponding to charge states 2+, 3+, and 4+, which were attributed to the successive loss of counterions

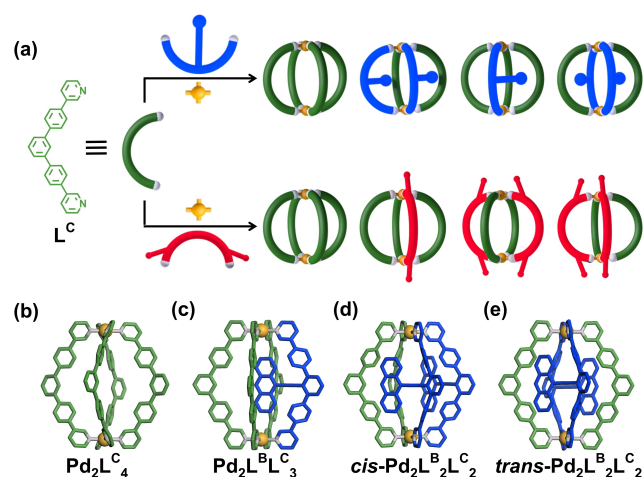


Figure 5. (a) The co-assembly of L^B and L^C at the top, and the co-assembly of L^A and L^C at the bottom. The crystal structure of (b) Pd_2L_4 , (d) $cis-Pd_2L_2L_2$ and (e) $trans-Pd_2L_2L_2$. (c) Pd_2L_3L have been simulated and structurally optimized using the Forcite tool as implemented in Materials Studio software (Hydrogen atoms are omitted for clarity).

BF_4^- . After deconvolution, the molecular weight of the assembly was determined to be 2552.66 Da, which is consistent with the expected heteroleptic cage $Pd_2L^A_2L^B_2$ (Figure 3e). Crystal structure analysis revealed a *trans*-configuration cage, named as *trans*- $Pd_2L^A_2L^B_2$ (Figure 3g). In *trans*- $Pd_2L^A_2L^B_2$, the bulky ethynyl anthracene groups on the two L^B ligands are positioned at the edges of the cage interior. This phenomenon is attributed to the flexibility of the acetylene bonds that allow for deflection and the multiple π - π interactions occurring between anthracene groups and benzene of adjacent molecular cages (Figure S76).

Furthermore, when one equivalent of L^B (dissolved in DMSO- d_6) was added to the CD_3CN solution of **Bowl**- $Pd_2L^A_3$ and stirred 8 h. The signals of **Bowl**- $Pd_2L^A_3$ completely disappeared in the 1H NMR spectrum, and a new set of proton peaks appeared (Figure 3d). The molecular composition of this new complex was determined by ESI-MS, named $Pd_2L^A_3L^B$ (Figure 3f). The crystal structure of $Pd_2L^A_3L^B$ also exhibited an offset of the bulky ethynyl anthracene groups towards the exterior of the cage (Figure 3h). However, unlike in *trans*- $Pd_2L^A_2L^B_2$, this offset is primarily due to the strong π - π interactions between anthracene groups of adjacent molecular cages (Figure S77). Meanwhile, the crystal structure provides a compelling explanation for the splitting of the methyl proton signals of the L^A in the 1H NMR spectrum: the outward offset of the ethynyl anthracene groups leads to differences in the chemical environments of the two opposing L^A ligands, resulting in the splitting of their proton signals into three distinct peaks in the spectrum (Figure 3d). More interestingly, when $Pd_2L^A_3L^B$ was left at room temperature for 60 days, it underwent a transformation into *trans*- $Pd_2L^A_2L^B_2$ (Figure S61). This discovery indicates that $Pd_2L^A_3L^B$ is

actually a kinetic product, which slowly converts into the thermodynamically more stable *trans*- $Pd_2L^A_2L^B_2$ over time.

In further research, we discovered that the assembly structures of **Ring**- $Pd_2L^A_2$, **Bowl**- $Pd_2L^A_3$, and **Cage**- $Pd_2L^A_4$ could undergo structural interconversion under specific conditions, such as the addition of metal or ligand L^A (Figures 4c, 4d). Based on this finding, we inferred that all three assembly structures could serve as intermediates to construct *trans*- $Pd_2L^A_2L^B_2$ or $Pd_2L^A_3L^B$. Such flexible intermediates could significantly enrich the synthesis pathways of the final heteroleptic cages. After conducting numerous experiments, we concluded that $Pd_2L^A_3L^B$ could be synthesized through as many as 10 different pathways, while *trans*- $Pd_2L^A_2L^B_2$ could be synthesized through 16 pathways.

For $Pd_2L^A_3L^B$, it cannot be synthesized in one pot (Figure 4a). When using the stepwise method shown in Figure 4c for synthesis, there are four different pathways originating from **Ring**- $Pd_2L^A_2$:

1. By adding the ligand L^A to **Ring**- $Pd_2L^A_2$, **Bowl**- $Pd_2L^A_3$ is formed. Subsequently, **Bowl**- $Pd_2L^A_3$ assembles with ligand L^B in a mixed solvent ($CD_3CN/DMSO-d_6$, v/v 2:1) at room temperature to generate $Pd_2L^A_3L^B$.
2. **Cage**- $Pd_2L^A_4$ is generated from **Ring**- $Pd_2L^A_2$, and then, when **Cage**- $Pd_2L^A_4$ reacts with the ligand L^B at 25 °C, it partially converts to $Pd_2L^A_3L^B$ with a conversion rate of approximately 47 % (Figure S67).
3. Firstly, **Cage**- $Pd_2L^A_4$ is generated from **Ring**- $Pd_2L^A_2$. Subsequently, **Bowl**- $Pd_2L^A_3$ is obtained from **Cage**- $Pd_2L^A_4$. Finally, **Bowl**- $Pd_2L^A_3$ assembles with the ligand L^B to produce $Pd_2L^A_3L^B$.
4. First, **Bowl**- $Pd_2L^A_3$ is generated from **Ring**- $Pd_2L^A_2$. Then, **Cage**- $Pd_2L^A_4$ is produced from **Bowl**- $Pd_2L^A_3$. Finally, when **Cage**- $Pd_2L^A_4$ reacts with the ligand L^B , it partially converts to $Pd_2L^A_3L^B$.

However, when starting from **Cage**- $Pd_2L^A_4$ or **Bowl**- $Pd_2L^A_3$, it is not possible to directly synthesize $Pd_2L^A_3L^B$ through **Ring**- $Pd_2L^A_2$. Therefore, there are only three ways to synthesize $Pd_2L^A_3L^B$ from **Cage**- $Pd_2L^A_4$ or **Bowl**- $Pd_2L^A_3$, as described in the Supporting Information (see the Supporting Information, Section 12). Thus, there are a total of 10 synthesis pathways for $Pd_2L^A_3L^B$.

For *trans*- $Pd_2L^A_2L^B_2$ (Figure 4d), starting from **Ring**- $Pd_2L^A_2$, there are five different pathways:

1. **Ring**- $Pd_2L^A_2$ directly assembles with ligand L^B in a mixed solvent ($CD_3CN/DMSO-d_6$, v/v 2:1) to generate *trans*- $Pd_2L^A_2L^B_2$.
2. Firstly, **Bowl**- $Pd_2L^A_3$ is generated from **Ring**- $Pd_2L^A_2$. Subsequently, **Bowl**- $Pd_2L^A_3$ assembles with ligand L^B at room temperature in CD_3CN to generate *trans*- $Pd_2L^A_2L^B_2$.
3. The **Cage**- $Pd_2L^A_4$ is generated from **Ring**- $Pd_2L^A_2$, and then partially converts to *trans*- $Pd_2L^A_2L^B_2$ with a conversion rate of approximately 50 % when **Cage**- $Pd_2L^A_4$ reacts with ligand L^B at 70 °C (Figure S68).
4. Firstly, **Cage**- $Pd_2L^A_4$ is generated from **Ring**- $Pd_2L^A_2$. Subsequently, **Bowl**- $Pd_2L^A_3$ is obtained from **Cage**- $Pd_2L^A_4$. Finally, **Bowl**- $Pd_2L^A_3$ assembles with ligand L^B to generate *trans*- $Pd_2L^A_2L^B_2$.

5. The **Bowl-Pd₂L^A₃** is generated from **Ring-Pd₂L^A₂**. Subsequently, **Cage-Pd₂L^A₄** is obtained from **Bowl-Pd₂L^A₃**. Finally, when **Cage-Pd₂L^A₄** reacts with the ligand **L^B**, it partially converts to **trans-Pd₂L^A₂L^B₂**.

Starting from **Cage-Pd₂L^A₄** or **Bowl-Pd₂L^A₃**, there are also five distinct pathways each for synthesizing **trans-Pd₂L^A₂L^B₂**. Additionally, **trans-Pd₂L^A₂L^B₂** can also be directly synthesized in a one-pot reaction using **L^A**, **L^B**, and Pd(II) (Figure 4b), resulting in a total of 16 synthesis routes. Taking the most complex pathway for synthesizing the two heteroleptic cages as an example: **L^A** and Pd(CH₃CN)₄(BF₄)₂ were assembled in a 1:1 equivalent ratio in CD₃CN for 8 hours to form **Ring-Pd₂L^A₂** (Figure 4e). To the **Ring-Pd₂L^A₂**, a DMSO-*d*₆ solution containing **L^A** was added at 70 °C and reacted for 8 hours to produce **Cage-Pd₂L^A₄** (Figure 4f). **Cage-Pd₂L^A₄** was precipitated from CD₃CN/DMSO-*d*₆ using diethyl ether and then reacted with a CD₃CN solution of Pd(CH₃CN)₄(BF₄)₂ at room temperature for 8 hours to form **Bowl-Pd₂L^A₃** (Figure 4g). Then, **Bowl-Pd₂L^A₃** reacted with **L^B** in CD₃CN at room temperature for 8 hours to form **trans-Pd₂L^A₂L^B₂** (Figure 4h). Alternatively, adding **L^B** (dissolved in DMSO-*d*₆) to the CD₃CN solution of **Bowl-Pd₂L^A₃** and reacting for 8 hours at room temperature can form **Pd₂L^A₃L^B** (Figure 4i).

Under the synergy of inner and outer steric hindrance, we successfully constructed two types of heteroleptic cages and achieved different pathway synthesis. To confirm IOSS strategy, we designed a control ligand, **L^C** (Figure 5a), which, due to the absence of both internal and external steric hindrance, selectively forms **Pd₂L^C₄** irrespective of the ratio of **L^C** to Pd(II) (Figures S28, S29). To investigate the effect of internal steric hindrance alone, **L^B** and **L^C** were employed for co-assembly with Pd(II). The presence of multiple assembly products in the system was indicated by the complex proton peak signals observed in the ¹H NMR spectrum (Figure S53). ESI-MS analysis revealed the molecular weights corresponding to three compounds: **Pd₂L^C₄**, **Pd₂L^B₃L^C₁** and **Pd₂L^B₂L^C₂** (Figure S54). Notably, for **Pd₂L^B₂L^C₂**, there may exist two configurations, *cis* and *trans*, which could not be distinguished solely by MS. Fortunately, through different crystal growth conditions, we were able to obtain crystals of **Pd₂L^C₄**, *cis*-**Pd₂L^B₂L^C₂**, and *trans*-**Pd₂L^B₂L^C₂** (Figure 5b, 5d, 5e), thereby confirming the existence of four compounds in the system. Furthermore, to assess the impact of external steric hindrance alone, we evaluated the co-assembly of **L^A**, **L^C**, and Pd(II). Regardless of whether a stepwise or one-pot method was employed, the co-assembly inevitably led to the formation of mixtures (Figures S46–S52). Through control experiments, we have fully confirmed the necessity of IOSS strategy. In the co-assembly system, it exhibits significant advantages in configuration control compared to single steric hindrance control strategies.

Conclusion

In summary, we conducted a detailed investigation about the pathway engineering and achieved the controlled

construction of the heteroleptic cages **Pd₂L^A₃L^B** and **trans-Pd₂L^A₂L^B₂** by IOSS strategy. Additionally, leveraging the flexible transformation among **Ring-Pd₂L^A₂**, **Bowl-Pd₂L^A₃**, and **Cage-Pd₂L^A₄**, we realized the pathway engineering of **Pd₂L^A₃L^B** and **trans-Pd₂L^A₂L^B₂**. Specifically, **Pd₂L^A₃L^B** can be constructed through 10 different pathways, while the pathways for constructing **trans-Pd₂L^A₂L^B₂** can reach up to 16, effectively. Through further control experiments, we demonstrated that using a single inner/outer steric hindrance strategy cannot effectively control the configuration of heteroleptic cages, leading to the formation of mixtures of various assemblies. This highlights the unique advantages of IOSS strategy in constructing complex structures. This work provides a reliable strategy for the controlled construction of heteroleptic cages, laying the foundation for further exploration of complex self-assembly systems and their applications in the future.

Supporting Information

Synthetic details and characterizations of ligands and metal-complexes including NMR, ESI–MS and Single-crystal data (PDF). The authors have cited additional references within the Supporting Information.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: Metal-Organic Cage • Pathway Engineering • Heteroleptic Cage • Supramolecular Chemistry • Single Crystal

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